

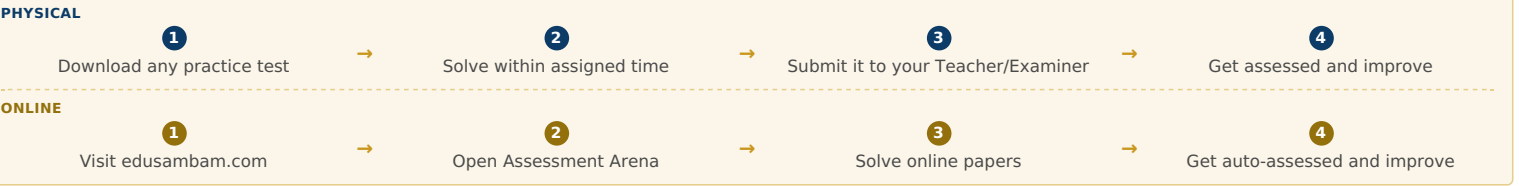
EDUSAMBAM ASSESSMENT SERIES

Chemistry Comprehensive Practice Test 5

PRINTABLE PRACTICE WORKSHEET

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YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP



Candidate Name:		Roll No.:	
Class / Grade:		Date:	
School / Institution:		Test Centre:	

Subject	Chemistry	Topic	Chemistry Comprehensive Practice Test 5	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. Which property is typical of metals?

Basic • 2 pts

- A.** Poor electrical conductivity
C. Malleability

B. Brittleness in all forms
D. Lack of lustre

Answer: ☐A ☐B ☐C ☐D

2. Which functional group identifies an alcohol?

Basic • 2 pts

- A.** -COOH
C. -CHO

B. -OH
D. -CO-

Answer: ☐A ☐B ☐C ☐D

3. At constant pressure, heating a liquid generally causes its particles to:

Basic • 2 pts

- A.** Lose all motion
C. Become neutrons

B. Gain kinetic energy
D. Stop colliding

Answer: ☐A ☐B ☐C ☐D

4. If 2 mol of H₂ react completely with O₂ according to 2H₂ + O₂ → 2H₂O, how many moles of H₂O form?

Basic • 2 pts

- A.** 1
C. 3

B. 2
D. 4

Answer: ☐A ☐B ☐C ☐D

5. Which statement best explains why catalysts speed many reactions?

Intermediate • 2 pts

- A.** They permanently increase product energy
C. They change the equilibrium constant in every reaction

B. They provide an alternative pathway with lower activation energy
D. They are consumed as reactants

Answer: ☐A ☐B ☐C ☐D

6. In a galvanic cell, chemical energy is primarily converted into: A. Electrical energy C. Nuclear energy Answer: ☐ A ☐ B ☐ C ☐ D	B. Sound energy D. Gravitational energy	Intermediate • 2 pts
7. Which pair is most likely to form a covalent molecular compound? A. Sodium and chlorine C. Carbon and oxygen Answer: ☐ A ☐ B ☐ C ☐ D	B. Magnesium and oxygen D. Potassium and bromine	Intermediate • 2 pts
8. Which particle has a positive charge? A. Electron C. Neutron Answer: ☐ A ☐ B ☐ C ☐ D	B. Proton D. Photon	Intermediate • 2 pts
9. The atomic number of an element equals the number of: A. Neutrons only C. Protons plus neutrons Answer: ☐ A ☐ B ☐ C ☐ D	B. Protons only D. Electrons plus neutrons	Intermediate • 2 pts
10. Which gas is a major product of complete combustion of a hydrocarbon? A. Carbon dioxide C. Hydrogen chloride Answer: ☐ A ☐ B ☐ C ☐ D	B. Sodium chloride D. Nitrogen metal	Intermediate • 2 pts
11. Which subatomic particle has approximately one atomic mass unit and no charge? A. Electron C. Neutron Answer: ☐ A ☐ B ☐ C ☐ D	B. Proton D. Positron	Intermediate • 2 pts
12. Which bond is typical between atoms of a metal lattice? A. Metallic C. Covalent only Answer: ☐ A ☐ B ☐ C ☐ D	B. Ionic only D. Hydrogen	Advanced • 2 pts
13. What does concentration describe in a solution? A. Amount of solute relative to a specified amount of solution or solvent C. Temperature only Answer: ☐ A ☐ B ☐ C ☐ D	B. Colour only D. Density of the container	Advanced • 2 pts
14. Which compound contains the –COOH functional group? A. Alkane C. Carboxylic acid Answer: ☐ A ☐ B ☐ C ☐ D	B. Alcohol D. Alkene	Advanced • 2 pts
15. If a reaction releases heat to its surroundings, the reaction is: A. Endothermic C. Neutral only Answer: ☐ A ☐ B ☐ C ☐ D	B. Exothermic D. Catalytic by definition	Advanced • 2 pts

Section B — True / False	20 marks
16. Acids can react with some metals to release hydrogen gas. A. True ☐ B. False ☐	Basic • 2 pts
17. The nucleus contains protons and neutrons. A. True ☐ B. False ☐	Basic • 2 pts
18. Metallic bonding involves attraction between positive metal ions and delocalized electrons. A. True ☐ B. False ☐	Basic • 2 pts
19. A mixture always has a fixed chemical formula. A. True ☐ B. False ☐	Basic • 2 pts
20. Distillation can separate liquids using differences in boiling points. A. True ☐ B. False ☐	Intermediate • 2 pts
21. Oxidation means gain of electrons in every accepted redox definition. A. True ☐ B. False ☐	Intermediate • 2 pts

22. Alcohols contain a hydroxyl functional group.
A. True ☐ **B.** False ☐

23. Changing the amount of a reactant can never affect reaction rate.
A. True ☐ **B.** False ☐

24. At equilibrium, reactant and product concentrations must be equal.
A. True ☐ **B.** False ☐

25. Corrosion is generally an electrochemical oxidation process.
A. True ☐ **B.** False ☐

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Section C — Fill in the Blank

20 marks

26. The smallest unit of an element that retains its chemical identity is an _____.
Answer: _____

27. The horizontal arrangement of elements by increasing atomic number is the periodic _____.
Answer: _____

28. A substance with a pH above 7 is generally _____.
Answer: _____

29. The change from solid to liquid is called _____.
Answer: _____

30. The material present in the larger amount in a solution is usually the _____.
Answer: _____

31. A reaction in which one compound breaks into simpler substances is a _____ reaction.
Answer: _____

32. A metal that loses electrons in a redox process is _____.
Answer: _____

33. The number of particles in one mole is Avogadro's _____.
Answer: _____

34. Compounds with the same molecular formula but different arrangements can be _____.
Answer: _____

35. A material that conducts electricity because of mobile ions is an _____.
Answer: _____

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Section D — Matching

10 marks

36. Match each term to its correct definition.

Column A	Answer	Column B
Nucleus	_____	A. Metallic bonding
Chromatography	_____	B. Carboxylic acid
Metallic bonding	_____	C. Electrolyte
Carboxylic acid	_____	D. Nucleus
Electrolyte	_____	E. Chromatography

Section E — Short Answer

20 marks

41. What is a hydrocarbon?

42. Why are noble gases relatively unreactive?

Advanced • 4 pts

Advanced • 4 pts

43. How can chromatography help identify components of a mixture?

Advanced · 4 pts

44. Explain the role of a salt bridge in a galvanic cell.

Advanced · 4 pts

45. Why can different structural arrangements produce different organic compounds with the same molecular formula?

Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: _____ / 100

Checked By: _____

Result / Grade: _____

Date Checked: _____

EDUSAMBAM ASSESSMENT SERIES

Chemistry Comprehensive Practice Test 5 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. C | 9. B |
| 2. B | 10. A |
| 3. B | 11. C |
| 4. B | 12. A |
| 5. B | 13. A |
| 6. A | 14. C |
| 7. C | 15. B |
| 8. B | |

Section B — True / False

- | | |
|-----------|-----------|
| 16. False | 21. True |
| 17. False | 22. False |
| 18. False | 23. True |
| 19. True | 24. True |
| 20. False | 25. False |

Section C — Fill in the Blank

- | | |
|-------------|-------------------|
| 26. atom | 31. decomposition |
| 27. table | 32. oxidized |
| 28. basic | 33. number |
| 29. melting | 34. isomers |
| 30. solvent | 35. electrolyte |

Section D — Matching

- | | |
|--------------------------|-------------------------|
| 36. Nucleus — D | 39. Carboxylic acid — B |
| 37. Chromatography — E | 40. Electrolyte — C |
| 38. Metallic bonding — A | |

Section E — Short Answer (Model Answers)

- compound containing only carbon and hydrogen
- outer electron shell relatively complete; little tendency to gain or lose electrons
- different substances move different distances/rates; compare spots/bands with standards
- maintains electrical neutrality; allows ion migration; completes internal circuit
- different connectivity of atoms; isomers can have different properties